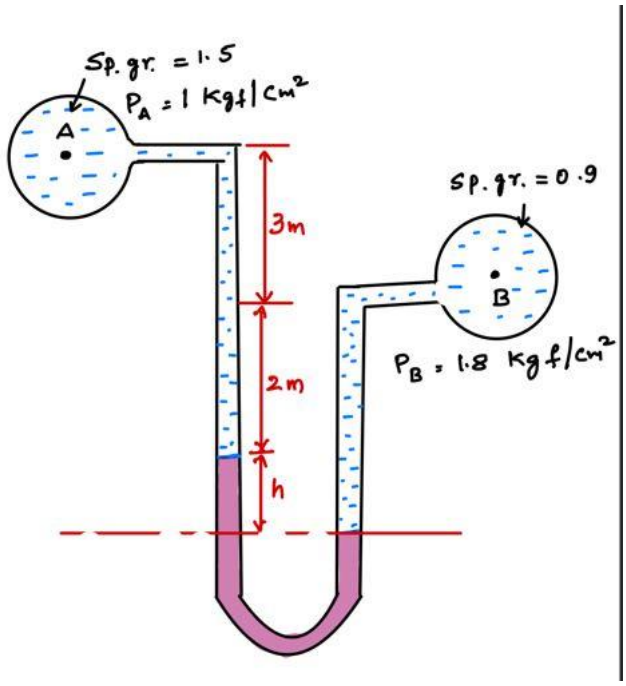
MANIPAL INSTITUTE OF TECHNOLOGY
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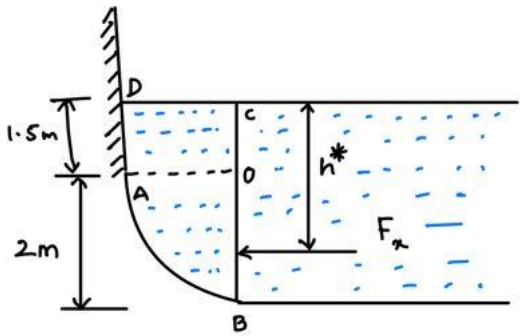
DEPARTMENT OF Aeronautical and Automobile Engineering			
END SEMESTER EXAMINATION		**NOV-DEC, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Automobile Engineering**	**Course Type**	**C**
Course Name	**Fluid Mechanics**	**Course Code**	**AAE 2128**
Semester	**III**	**Max Marks**	**50**
Date	**27-11-2024**	**Duration**	**3 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A	Two large plane surfaces are 2.4 cm apart. The space between the surfaces is filled with glycerine. What force is required to drag a very thin plate of surface area 0.5 square meters between the two large plane surfaces at a speed of 0.6 m/s, if: (i) The thin plate is in the middle of the two plane surfaces, and (ii) The thin plate is at a distance of 0.8 cm from one of the plane surfaces? Take the dynamic viscosity of glycerine = 8.10 x 10^{-1} Ns/m^2	**4**	1	1, 3, 7	**1**	**1,2**	**3**
1B	Derive Pascal's law and discuss the physical meaning of the mathematical expression obtained.	**3**	1	1, 2, 7	**1**	**1,2**	**2**
1C	A differential manometer is connected at the two points A and B of two pipes as shown in the figure. The pipe A contains a liquid of specific gravity = 1.5 while pipe B contains a liquid of specific gravity = 0.9. The pressures at A and B are 1 kgf/cm^2 and 1.80 kgf/cm^2 respectively. Find	**3**	1	1, 2, 7	**1**	**1,2**	**3**

	the difference in mercury level in the differential manometer in mm.  						
2A	Define the following: (i) Streak line (ii) velocity potential function	**2**	2	1, 2, 7	**2**	**1,2**	**1**
2B	Derive the three-dimensional continuity equation for steady incompressible flow in a cartesian coordinate system.	**4**	2	1, 2, 7	**2**	**1,2**	**2**
2C	The velocity in a fluid flow is given $V= 4x^3i-10x^2yj+2tk$. Find the velocity and acceleration of a fluid particle at (2, 1, 3) at time t=1.	**4**	2	1, 2, 7	**2**	**1,2**	**3**
3A	Water flows through pipe AB 1.2 m diameter at 3 m/s and then passes through a pipe BC 1.5 m diameter. At C, the pipe branches. Branch CD is 0.8 m in diameter and carries one-third of the flow in AB. The flow velocity in branch CE is 2.5 m/s. Find the volume rate of flow in AB, the velocity in BC, the velocity in CD, and the diameter of CE.	**3**	2	1, 2, 7	**2**	**1,2**	**3**
3B	Find the volume of water displaced and position of center of buoyancy for a wooden block of width 2.5 m and depth	**3**	3	1, 2, 7	**3**	**1,2**	**3**

	1.5 m, when it floats horizontally in the water. The density of the wooden block is 650 kg/m^3 and its length is 6 m.						
3c	A solid cylinder of diameter 4 m has a height of 4 m. Find the meta-centric height of the cylinder if the specific gravity of the material of the cylinder = 0.6 and it is floating in water with its axis vertical. State whether the equilibrium is stable or unstable.	**4**	3	1, 2, 7	**3**	**1,2**	**3**
4A	A horizontal venturimeter with an inlet diameter of 20 cm and throat diameter of 10 cm is used to measure the flow of water. The pressure at the inlet is 17.658 N/cm^2 and the vacuum pressure at the throat is 30 cm of mercury. Find the discharge of water through the venturimeter. consider C_d= 0.98.	**3**	4	1, 2, 7	**4**	**1,2**	**3**
4B	Derive the Eulers equation stating the assumptions made and arrive at Bernoulli's equation.	**4**	4	1, 2, 7	**4**	**1,2**	**2**
4C	Compute the horizontal and vertical components of the total force acting on a curved surface AB, which is in the form of a quadrant of a circle of radius 2 m as shown in the figure. Also, find the point of application of force in the horizontal direction. Consider the width of the gate to be unity. 	**3**	3	1, 2, 7	**3**	**1,2**	**3**
5A	The pressure difference Δp in a pipe of diameter D and length l due to turbulent flow depends on the velocity V, viscosity μ, density ρ, and roughness *k*. Using Buckinghams π theorem, show that $\frac{\Delta p}{\rho V^2} = f\left[\frac{1}{D}, \frac{\mu}{DV\rho}, \frac{k}{D}\right]$	**4**	4	1, 2, 7	**4**	**1,2**	**3**

	Consider D V and ρ as repeating variables.						
5B	With a neat diagram explain the development of a boundary layer over a plate and discuss how it is different in comparison to a fully developed, steady, incompressible flow inside a pipe.	**4**	5	1, 2, 7	**5**	**1,2**	**2**
5C	Discuss on 3 types of methods used to understand boundary layer thickness analytically.	**2**	5	1, 2, 7	**5**	**1,2**	**2**

Retain only one. Delete the others

xxxxxxx Fill in appropriately.

**** Applicable for only IET-accredited courses (modules). The IET columns may be deleted for non-IET courses.***

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***10***
2	***13***
3	***10***
4	***11***
5	***6***

BL	Total Marks
2	**21**
3	**27**
4	
5	